

LECTURE 4: MATH1061/7861
TOPICS: CONDITIONAL STATEMENTS AND THEIR
NEGATIONS
DATE: TUESDAY MARCH 3, 2026

Learning objectives:

(1) Understand the logical connectives $\rightarrow$ ('if ... then ...') and $\leftrightarrow$ ('... if and only if ...')

(2) Be familiar with logical equivalences involving $\rightarrow$:

$$\text{def} \quad \underline{p \rightarrow q} \equiv \sim p \vee q \equiv \underline{\sim q \rightarrow \sim p.}$$

Contrapositive

(3) Be able to negate a conditional statement.

(4) Understand the meaning of necessary and sufficient conditions.

There are many ways of expressing $p \rightarrow q$ in English:

(1) p implies q

(2) if p then q

(3) q if p

(4) p is sufficient for q

(5) p only if q

(6) q is necessary for p

Source of
Confusion:

Negation of
 $\underline{p \rightarrow q.}$

Q1: Consider the statement:

"If the file is not found, then the program crashes."

Which of the following has the same meaning as our statement?

~~Yes~~

(1) If the program crashes, then the file is not found.

Program can crash for other reasons

No

(2) If the file is found, then the program doesn't crash.

No

(3) The program crashes only if a file is not found.

No.

(4) If the program does not crash, then the file is not missing.

Yes

Contrapositive.

Positive integers 1, 2, 3, 4, ...

↓

Q2: Let $n \in \mathbb{Z}_{>0}$. Find conditions that are:

≡

(1) Necessary but not sufficient for n to be a multiple of 10.

n is divisible by 5.

n is even

$n \geq 10$

(2) Sufficient but not necessary for n to be a multiple of 10.

n is a multiple of 20

n is a power of 10. (i.e. $n = 10^k$ for some $k \in \mathbb{Z}_{>0}$).

(3) Necessary and sufficient for n to be a multiple of 10.

Last digit of n is 0.

n is a multiple of 2 & 5.

Q3: True or false:

$$\sim (p \rightarrow q) \equiv \sim p \rightarrow \sim q \quad \text{False.}$$

As noted in the videos / textbook
logical definition of $p \rightarrow q$ is $\sim p \vee q$.

This means $\sim (p \rightarrow q)$ is $\sim (\sim p \vee q)$

(This is negation of $p \rightarrow q$) $\xrightarrow{\text{De Morgan}}$ $\equiv$ $p \wedge \sim q$

On the other hand $\sim p \rightarrow \sim q$ is $\sim(\sim p) \vee (\sim q)$

So we want to know $\equiv$ $p \vee \sim q$

$p \wedge \sim q \equiv p \vee \sim q$ No.

p is T & q is F

$p \wedge \sim q$ is true & $p \vee \sim q$ is true

This doesn't help. Showing they are not equivalent

p & q both true.

LHS is F whereas RHS is True.